Paarth Bhasin

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INTRODUCTION

A customer-obsessed **Software Engineer** with 7+ years of experience delivering production software across distributed systems, cloud infrastructure, AI integrations and customer-facing applications. Combines deep systems-level problem-solving with hands-on full-stack development, API integration, automated testing and production observability.

I am a skilled builder with a unique insight into products and customers, that enables me to build world-class Software systems and products.

EXPERIENCE

▷ **Senior Full Stack Engineer (Financial, Rewards & Accounting)** Nov 2024 - Present
Big WX / Woolworths Group (https://bigw.com.au) *Sydney, NSW*
Responsibilities:

- Own the design and delivery of customer-facing e-commerce capabilities across checkout, payments, rewards and loyalty, spanning frontend applications and backend integrations serving 1M+ members weekly.
- Strengthened production observability through OpenTelemetry and Sentry, and expanded automated end-to-end testing of critical customer journeys using Cypress and Playwright.

Achievements:

- Increased cart checkout rates from 56% to 70% for customers with a loyalty card.
- Architected real-time asynchronous data pipelines to stream analytics data directly to Google Cloud (**GTM**), eliminating downstream microservice dependencies and resolving system redundancies.
- Led implementation of **Apple Pay**, **PayPal** and **Afterpay** across checkout, delivering secure payment integrations & authentication flows for a customer-facing commerce experience responsible for $15M+ in sales. Involved direct collaboration with customers, BAs, product managers, marketing executives & Director/CTO.
- Increased end to end test coverage of critical user walkthrough scenarios (eg. cart page, checkout etc.) in front-end, following TDD using **Cypress** and **PlayWright**.

▷ **Full Stack Engineer** Jun 2024 - Nov 2024
Evermed (https://evermed.com.au) *Lay-off due to startup financial problems at that time* *Sydney, NSW*
Responsibilities:

- Part of a 4 person startup, composed of ex-Google, ex-Rokt leadership, empowering customers across Australia, effectively acting in the capacity of a tech-lead
- Rollout of new e-commerce web & back-end features, optimizing existing features, reducing tech-debt and improving operations in a fast-paced environment, with lots of pair-programming.
- Working with **AWS**, **Python**, **C#/.NET**, **CSS**, **Node.js**, **React.js**, **Material UI**, **Nginx** for full-stack application development. Introduced test based approach cloud-infra and front-end applications, outlining testing strategy. Also performed user experience research.

Achievements:

- Significantly improved the front-end admin portal for pharmacy staff (eg. Priceline), to produce much more detailed information about customers and prescriptions, using **CSS** best-practices. Directly interacted with pharmacy staff to find areas of improvement.
- Presented an alternative to subsection of internal micro-services design to instead use **GraphQL**.

▷ **Research Engineer - Front End / AI** Apr 2024 - Jun 2024
MapAI (https://mapai.net) *Sydney, NSW*
Responsibilities:

- Built full-stack capabilities for geospatial analysis tools in collaboration with data scientists.

- Contributed to LLM integration using OpenAI and LangChain.
- Configured access to a university-provisioned GPU environment through AWS VPC networking for machine-learning workloads.

Achievements:

- Built a hoverable chat experience, that could run queries and keep history. Contributed to the LLM model integration with OpenAI, using **Python LangChain**.

▷ **Senior Software Engineer - Platform/SRE/Performance** Jan 2019 - Mar 2024
Atlassian *Sydney, NSW*
Responsibilities:

- Make Jira reliable by introducing key improvements in the application (**cache**, **DB optimization**, **algorithmic improvements**, **traffic proxying**, **rate-limiting**, **OS settings**), whilst maintaining 99.9% uptime SLA.
- Onboarded various team members to on-call, to help establish a new team based in Sydney. Mentor junior engineers through projects to be setup for success in leading initiatives.
- Participated in regular on-call roster for troubleshooting and incident management for Jira in AWS cloud, resolving customer concerns and diagnosing performance issues.
- Led project to define ownership of monitoring and key operational knowledge elements within the team, to have it fully functional and ready to operate as a dedicated SRE team.
- Managed all aspects of **CI/CD** pipelines - containerisation, deployments, and configuration as code in **Terraform**. This involved large scale scripting in **Terraform**, using best-practices like modules, remote state management, variables, version-management (**tfenv**), loops/counters.

Key Achievements:

- Led initiative for performance optimization of a widely used user-facing endpoint serving millions of customers, by *algorithm redesign* & **SQL** query optimization, improving endpoint latency by 50% on avg, eg **91s→35s**. Involved collaboration across different geographies. Utilized **AWS Kinesis** streams for long term data analysis.
- Developed a UI, used for monitoring, tracing & management of deployments for synthetic tests, using **GraphQL** (using **Graphene** based **Python** GraphQL server), **React Relay** & **TypeScript**. It is used by 50+ dev teams. This involved surveying engineering teams, to find customer pain-points, and prioritizing key features of the UI.
- Drove **OpenTelemetry** adoption in Jira in **Java**, to adopt observability best-practices and for improving the tracing footprint. This involved evaluation of third-party tools as well such as **Grafana**, **NewRelic**, **Honeycomb.io**, **Prometheus**. This also involved *open source* contributions to OpenTelemetry's Java framework.
- Tech lead for improving Jira's Message processing layer (AWS SQS) (**Event Driven Architecture**), introducing key improvements directly at the platform (**AWS**) and application layer (**resiliency algorithms**).
- Led an initiative to improve resiliency of database towards cluster fail-over to reduce customer downtime, requiring OS level programming related with **JDBC** database driver and **TCP/IP** layer network optimization to reduce time to recover (TTR), for (**Aurora**) across the fleet in multiple regions/data-centers. This reduced TTR from 30-60m to 45-60s on average and <1s for best case scenarios.
- Devised testing framework for **performance analysis** to unblock migration of large customers moving from server to cloud, for multiple feature teams, (eg. Equifax - with 50k+ users) to adopt our cloud services. This developed an underp-pinning strategy & principles for reliability, performance & scale testing strategy to conduct such testing for entire Jira Dev org, in various geographies (∼1500-2000 developers). Participated in deep stakeholder requirements analysis and problem solving, taking the stakeholders through the journey end-to-end.
- Developed automation (**Python**) for auto-scaling of **WebServer (AWS EC2)**, to handle to high tomcat thread loads in Jira, improving downtime during peak traffic, and TTR to <5m, TTD < 5s and uptime at 99.9% SLA.

EDUCATION

Masters of Computer Engineering, *University of New South Wales - Distinction, WAM: 71.0* 2019 - 2022
(Part-time)
Bachelor of Software Engineering, *Monash University* 2014 - 2018